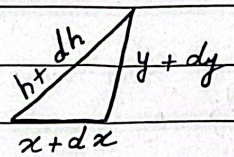
Ex # 13.4

Date:

Total Differential :-

$$z = z(x, y)$$



$$dz = \frac{dz}{dx}dx + \frac{\partial z}{\partial y}dy$$

$$F = F(x_1, x_2, x_3, x_4)$$

$$dF = \frac{\partial F}{\partial x_1}dx_1 + \frac{\partial F}{\partial x_2}dx_2 + \frac{\partial F}{\partial x_3}dx_3 + \frac{\partial F}{\partial x_4}dx_4$$

9-20 Compute the differential dz or dw of the function.

9. $z = 7x - 2y$

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(7x - 2y) = 7, \quad \frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(7x - 2y) = -2$$

$$dz = (7)dx + (-2)dy$$
$$= 7\,dx - 2\,dy$$

10. $z = e^{xy}$

$$\frac{\partial z}{\partial x} = e^{xy} \cdot y = ye^{xy}, \quad \frac{\partial z}{\partial y} = e^{xy} \cdot x = xe^{xy}$$

$$dz = ye^{xy}\,dx + xe^{xy}\,dy$$

11. $z = x^3y^2$

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(x^3y^2) = 3x^2y^2, \quad \frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(x^3y^2) = 2x^3y$$

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$$dz = 3x^2y^2\,dx + 2x^3y\,dy$$ ans

12. $z = 5x^2y^5 - 2x + 4y + 7$

solution

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}(5x^2y^5 - 2x + 4y + 7) = 10xy^5 - 2$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(5x^2y^5 - 2x + 4y + 7) = 25x^2y^4 + 4$$

$$dz = (10xy^5 - 2)\,dx + (25x^2y^4 + 4)\,dy$$ ans

13. $z = \tan^{-1} xy$

solution

$$\frac{\partial z}{\partial x} = \frac{1}{1+(xy)^2} \cdot \frac{\partial}{\partial x}(xy) = \frac{y}{1+x^2y^2}$$

$$\frac{\partial z}{\partial y} = \frac{1}{1+(xy)^2} \cdot \frac{\partial}{\partial y}(xy) = \frac{x}{1+x^2y^2}$$

$$dz = \frac{y}{1+x^2y^2}\,dx + \frac{x}{1+x^2y^2}\,dy$$ ans

14. $z = e^{-3x}\cos 6y$

solution

$$\frac{\partial z}{\partial x} = \cos 6y \cdot \frac{\partial}{\partial x}(e^{-3x}) = -3e^{-3x}\cos 6y$$

$$\frac{\partial z}{\partial y} = e^{-3x} \cdot \frac{\partial}{\partial y}(\cos 6y) = -6e^{-3x}\sin 6y$$

$$dz = -3e^{-3x}\cos(6y)\,dx - 6e^{-3x}\sin(6y)\,dy$$ ans

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15. $w = 8x - 3y + 4z$

$\frac{\partial w}{\partial x} = 8$, $\frac{\partial w}{\partial y} = -3$, $\frac{\partial w}{\partial z} = 4$

$dw = 8\,dx - 3\,dy + 4\,dz$

16. $w = e^{xyz}$

$\frac{\partial w}{\partial x} = yz\,e^{xyz}$, $\frac{\partial w}{\partial y} = xz\,e^{xyz}$, $\frac{\partial w}{\partial z} = xy\,e^{xyz}$

$dw = yz\,e^{xyz}\,dx + xz\,e^{xyz}\,dy + xy\,e^{xyz}\,dz$

17. $w = x^3y^2z$

$\frac{\partial w}{\partial x} = 3x^2y^2z$, $\frac{\partial w}{\partial y} = 2x^3yz$, $\frac{\partial w}{\partial z} = x^3y^2$

$dw = 3x^2y^2z\,dx + 2x^3yz\,dy + x^3y^2\,dz$

18. $w = 4x^2y^3z^7 - 3xy + z + 5$

$\frac{\partial w}{\partial x} = 8xy^3z^7 - 3y$, $\frac{\partial w}{\partial y} = 12x^2y^2z^7 - 3x$, $\frac{\partial w}{\partial z} = 28x^2y^3z^6 + 1$

$dw = (8xy^3z^7 - 3y)\,dx + (12x^2y^2z^7 - 3x)\,dy + (28x^2y^3z^6 + 1)\,dz$

19. $w = \tan^{-1}(xyz)$

$\frac{\partial w}{\partial x} = \frac{yz}{1+x^2y^2z^2}$, $\frac{\partial w}{\partial y} = \frac{xz}{1+x^2y^2z^2}$, $\frac{\partial w}{\partial z} = \frac{xy}{1+x^2y^2z^2}$

$dw = \frac{yz}{1+x^2y^2z^2}\,dx + \frac{xz}{1+x^2y^2z^2}\,dy + \frac{xy}{1+x^2y^2z^2}\,dz$

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20. $w = \sqrt{x} + \sqrt{y} + \sqrt{z}$

Solution

$$dw = \frac{1}{2\sqrt{x}}dx + \frac{1}{2\sqrt{y}}dy + \frac{1}{2\sqrt{x}}dz$$

Ans

26 Use a total differential to approximate the change in the values of f from P to Q. Compare your estimate with the actual change in f.

31. $f(x,y) = x^2 + 2xy - 4x$; $P(1,2)$, $Q(1.01, 2.04)$

Solution

1) Find the partial derivatives.

$$f_x(x,y) = 2x + 2y - 4$$

$$f_y(x,y) = 2x$$

2) Evaluate the partial derivatives at point $P(1,2)$.

$$f_x(1,2) = 2(1) + 2(2) - 4 = 2$$
$$f_y(1,2) = 2(1) = 2$$

3) Find the changes dx and dy:

$$dx = x_Q - x_P = 1.01 - 1 = 0.01$$
$$dy = y_Q - y_P = 2.04 - 2 = 0.04$$

4) Calculate the total differential df:

$$df = f_x(1,2)\,dx + f_y(1,2)\,dy$$
$$= (2)(0.01) + (2)(0.04) = 0.10$$

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- Value at initial point $P(1,2)$.

$$f(P) = f(1,2) = (1)^2 + 2(1)(2) - 4(1) = 1+4-4 = 1$$

- Approximate value at Q.

$$\text{Approximate Value} = f(P) + df$$
$$= 1 + 0.10$$
$$= 1.10$$

- Find the actual value at Q.

$$f(Q) = f(1.01, 2.04) = (1.01)^2 + 2(1.01)(2.04) - 4(1.01)$$
$$= 1.0201 + 4.1208 - 4.04$$
$$= 1.1009$$

- Find the error.

$$\text{Error} = \text{Actual Value} - \text{Approximate Value}$$
$$= 1.1009 - 1.10$$
$$= 0.0009$$

22. $f(x,y) = x^{\frac{1}{3}} y^{\frac{1}{2}}$; $P(8,9)$, $Q(7.78, 9.03)$

Solve

- First we calculate the value of the function at starting point $P(8,9)$.

$$f(8,9) = (8)^{\frac{1}{3}} (9)^{\frac{1}{2}} = 2.3 = 6$$

- Find the changes dx and dy:

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$dx = x_Q - x_P = 7.78 - 8 = -0.22$

$dy = y_Q - y_P = 9.03 - 9 = 0.03$

- Find the partial derivatives.

$$f_x(x,y) = \frac{1}{3} x^{-2/3} y^{1/2}, \quad f_y(x,y) = \frac{1}{2} x^{\frac{1}{3}} y^{-\frac{1}{2}}$$

- Now, we evaluate these derivatives at the point $P(8,9)$.

$$f_x(8,9) = \frac{1}{3}(8)^{-2/3}(9)^{\frac{1}{2}} = \frac{1}{4}$$

$$f_y(8,9) = \frac{1}{2}(8)^{\frac{1}{3}}(9)^{-\frac{1}{2}} = \frac{1}{3}$$

- Calculate the total differential df:

$$df = f_x(P)\,dx + f_y(P)\,dy$$

$$df = \left(\frac{1}{4}\right)(-0.22) + \left(\frac{1}{3}\right)(0.03) = -0.045$$

- Approximate value at Q.

$$f(Q) \approx f(P) + df$$
$$\approx 6 + (-0.45)$$

Approximate Value = 5.955

- Actual Value at Q:

$$f(Q) = f(7.78, 9.03) = (7.78)^{\frac{1}{3}} \cdot (9.03)^{\frac{1}{3}}$$

$$\approx 5.954387$$

Error = 5.954387 − 5.955

Error = −0.000613

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$$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$$

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33-40 a) Find the local linear approximation L to the specified function f at the designated point P.
b) Compare the error in approximating f by L at the specified point Q with the distance between P and Q.

33. $f(x,y) = \dfrac{1}{\sqrt{x^2+y^2}}$; $P(4,3)$, $Q(3.92, 3.01)$

Solve

Find the function value at $P(4,3)$.

$$f(4,3) = \frac{1}{\sqrt{4^2+3^2}} = \frac{1}{\sqrt{25}} = \frac{1}{5}$$

Find the partial derivatives.

$$f_x = -\frac{1}{2}(x^2+y^2)^{-3/2}(2x) = \frac{-x}{(x^2+y^2)^{3/2}}$$

$$f_y = -\frac{1}{2}(x^2+y^2)^{-3/2}(2y) = \frac{-y}{(x^2+y^2)^{3/2}}$$

Evaluate the partial derivatives at $P(4,3)$.

$$f_x(4,3) = \frac{-4}{(4^2+3^2)^{3/2}} = \frac{-4}{(25)^{3/2}} = \frac{-4}{125}$$

$$f_y(4,3) = \frac{-3}{(4^2+3^2)^{3/2}} = \frac{-3}{(25)^{3/2}} = \frac{-3}{125}$$

Construct the linear approximation $L(x,y)$.

$$L(x,y) = \frac{1}{5} + \left(\frac{-4}{125}\right)(x-4) + \left(\frac{-3}{125}\right)(y-3)$$

$$L(x,y) = \frac{1}{5} - \frac{4}{125}(x-4) - \frac{3}{125}(y-3)$$

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- Approximation Value $L(Q)$:

$$L(3.92, 3.01) = \frac{1}{5} - \frac{4}{125}(3.92-4) - \frac{3}{125}(3.01-3)$$

$$= 0.2 + \frac{0.32}{125} - \frac{0.03}{125} = 0.20232$$

- Actual Value $f(Q)$:

$$f(3.92, 3.01) = \frac{1}{\sqrt{(3.92)^2 + (3.01)^2}} \approx 0.20233$$

- Error $= |f(Q) - L(Q)|$

$$= |0.20233 - 0.20232|$$
$$\approx 0.0000141$$

- Calculate the distance between P and Q.

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(3.92-4)^2 + (3.01-3)^2}$$

$$= \sqrt{0.0064 + 0.0001}$$
$$= \sqrt{0.0065}$$

$$\approx 0.08062$$

34. $f(x,y) = x^{0.5} y^{0.3}$; $P(1,1)$, $Q(1.05, 0.97)$

a) Linear Approximation L.

- $f(1,1) = 1$

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$f_x = 0.5x^{-0.5}y^{0.3} \Rightarrow f_x(1,1) = 0.5$

$f_y = 0.3x^{0.5}y^{-0.7} \Rightarrow f_y(1,1) = 0.3$

$L(x,y) = 1 + 0.5(x-1) + 0.3(y-1)$ done

b) Error and Distance.

- Approximate Value: $L(1.05, 0.97) = 1 + 0.5(0.05) + 0.3(-0.03)$
  $= 1.016$

- Actual Value: $f(1.05, 0.97) = (1.05)^{0.5}(0.97)^{0.3}$
  $= 1.015937$

- Error $= |1.015937 - 1.016| \approx 0.000063$.

- Distance $|PQ| = \sqrt{(0.05)^2 + (-0.03)^2} \approx 0.05831$

35. $f(x,y) = x \sin y$; $P(0,0)$, $Q(0.003, 0.004)$. done

a) Linear Approximation L.

$f(0,0) = 0$

$f_x = \sin y \Rightarrow f_x(0,0) = 0$

$f_y = x\cos y \Rightarrow f_y(0,0) = 0$

$L(x,y) = 0$. done

b) Error and Distance.

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Approximate Value : $L(0.003, 0.004) = 0$

Actual Value : $f(0.003, 0.004) = 0.003 \sin(0.004)$
≈ 0.000012

Error $= |0.000012 - 0| = 0.000012$

Distance $|PQ| = \sqrt{(0.003)^2 + (0.004)^2} = 0.005$

36. $f(x,y) = \ln(xy)$; $P(1,2)$, $Q(1.01, 2.02)$

a) Linear Approximation L.

$f(1,2) = \ln(2)$

$f_x = \frac{1 \cdot y}{xy} = \frac{1}{x} \Rightarrow f_x(1,2) = 1$

$f_y = \frac{1}{xy} \cdot x = \frac{1}{y} \Rightarrow f_y(1,2) = \frac{1}{2}$

$L(x,y) = \ln(2) + 1(x-1) + \frac{1}{2}(y-2)$.

b) Error and Distance.

Approximate Value : $L(1.01, 2.02) = \ln(2) + (0.01) + \frac{1}{2}(0.02)$
$= \ln(2) + 0.02$
≈ 0.713147

Actual Value : $f(1.01, 2.02) = \ln(2.0402) \approx 0.713048$

Error $= |0.713048 - 0.713147| \approx 0.000099$

Distance $|PQ| = \sqrt{(0.01)^2 + (0.02)^2} \approx 0.02236$

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37. $f(x,y,z) = xyz$; $P(1,2,3)$, $Q(1.001, 2.002, 3.003)$

a) Linear Approximation L.

$f(1,2,3) = 6$

$f_x = yz \Rightarrow f_x(P) = 6$

$f_y = xz \Rightarrow f_y(P) = 3$

$f_z = xy \Rightarrow f_z(P) = 2$

$L(x,y,z) = 6 + 6(x-1) + 3(y-2) + 2(z-3)$.

b) Error and Distance.

Approximate Value: $L(Q) = 6 + 6(0.001) + 3(0.002) + 2(0.003)$
$= 6.018$

Actual Value: $f(Q) = (1.001)(2.002)(3.003) \approx 6.018018$

Error $= |6.018018 - 6.018| = 0.000018$

Distance $|PQ| = \sqrt{(0.001)^2 + (0.002)^2 + (0.003)^2}$
≈ 0.00374

38. $f(x,y,z) = \dfrac{x+y}{y+z}$; $P(-1,1,1)$, $Q(-0.99, 0.99, 1.01)$

a) Linear Approximation L.

$f(-1,1,1) = \dfrac{-1+1}{1+1} = 0$

$f_x = \dfrac{1}{y+z} \Rightarrow f_x(P) = \dfrac{1}{1+1} = \dfrac{1}{2}$

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$$f_y = \frac{(1)(y+z) - (x+y)(1)}{(y+z)^2} = \frac{z-x}{(y+z)^2}$$

$$f_y(P) = \frac{1-(-1)}{(2)^2} = \frac{2}{4} = \frac{1}{2}$$

$$f_z = \frac{-(x+y)}{(y+z)^2} \Rightarrow f_z(P) = \frac{-(-1+1)}{(2)^2} = 0$$

$$L(x,y,z) = 0 + \frac{1}{2}(x+1) + \frac{1}{2}(y-1) + 0(z-1)$$

$$= \frac{1}{2}(x+1) + \frac{1}{2}(y-1).$$

b) Error and Distance.

Approximation Value : $L(Q) = \frac{1}{2}(-0.99+1) + \frac{1}{2}(0.99-1) + \frac{1}{2}(-0.01).$

$= 0.$

Error $= |0-0| = 0$

Distance $|PQ| = \sqrt{(0.01)^2 + (-0.01)^2 + (0.01)^2}$

$\approx 0.01732.$

49-52 A function f is given along with a local linear approximation L to f at a point P. Use the information given to determine point P.

49. $f(x,y) = x^2 + y^2 ; L(x,y) = 2y - 2x - 2$

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53. The length and width of a rectangle are measured with errors of at most 3% and 5%, respectively. Use differentials to approximate the maximum percentage error in the calculated area.

Solve

$$A = xy$$

$$\% \text{ error in } x = 3\% = \frac{dx}{x} \times 100$$

$$\% \text{ error in } y = 5\% = \frac{dy}{y} \times 100$$

$$\% \text{ error in } A = \frac{dA}{A} \times 100 = ?$$

$$dA = \frac{\partial A}{\partial x}\,dx + \frac{\partial A}{\partial y}\,dy$$

$$dA = y\,dx + x\,dy$$

$$\frac{dA}{A} = \frac{y\,dx}{xy} + \frac{x\,dy}{xy}$$

$$\frac{dA}{A} = \frac{dx}{x} + \frac{dy}{y}$$

$$\frac{dA}{A} \times 100 = \frac{dx}{x} \times 100 + \frac{dy}{y} \times 100$$

$$\text{Error} = 3\% + 5\%$$

$$= 8\%$$

Ans

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54. The radius and height of a right circular cone are measured with error of most 1% and 4%, respectively. Use differentials to approximate the maximum percentage error in the calculated volume.

Solution

. Error in radius $= \frac{dr}{r} \times 100 = 1\%$

. Error in height $= \frac{dh}{h} \times 100 = 4\%$

. Maximum % error in the volume (V), which is

$$\frac{dV}{V} \times 100 = ?$$

Function: $V = \frac{1}{3}\pi r^2 h$

$$dV = \frac{\partial V}{\partial r}dr + \frac{\partial V}{\partial h}dh$$

$$dV = \left(\frac{2}{3}\pi rh\right)dr + \left(\frac{1}{3}\pi r^2\right)dh$$

$$\frac{dV}{V} = \frac{\left(\frac{2}{3}\pi rh\right)dr}{\frac{1}{3}\pi r^2 h} + \frac{\left(\frac{1}{3}\pi r^2\right)dh}{\frac{1}{3}\pi r^2 h} = 2\left(\frac{dr}{r}\right) + \frac{dh}{h}$$

Multiply by 100: $\frac{dV}{V} \times 100 = 2\left(\frac{dr}{r} \times 100\right) + \left(\frac{dh}{h} \times 100\right)$

$$\frac{dV}{V} \times 100 = 2(1\%) + (4\%)$$

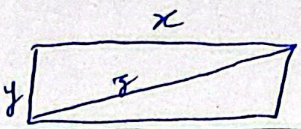
$$= 2\% + 4\% = 6\%$$

The maximum percentage error in the calculated volume is 6%.

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55. The length and width of a rectangle are measured with errors of most r%, where r is small. Use differentials to approximate the ~~minimum~~ maximum percentage error in the calculated length of the diagonal.

Solution

$\%$ Error in length $= \frac{dx}{x} \times 100 = r\%$

$\%$ Error in width $= \frac{dy}{y} \times 100 = r\%$

$\%$ Error in diagonal's length $(z) = \frac{dz}{z} \times 100 = ?$

Function : $z^2 = x^2 + y^2$

$$2z\,dz = 2x\,dx + 2y\,dy$$
$$z\,dz = x\,dx + y\,dy$$

$$\frac{z\,dz}{z^2} = \frac{x\,dx}{x^2+y^2} + \frac{y\,dy}{x^2+y^2}$$

$$\frac{dz}{z} = \frac{x^2}{x^2+y^2}\left(\frac{dx}{x}\right) + \frac{y^2}{x^2+y^2}\left(\frac{dy}{y}\right)$$

$$= \frac{x^2}{x^2+y^2}(r\%) + \frac{y^2}{x^2+y^2}(r\%)$$

$$= r\%\left(\frac{x^2+y^2}{x^2+y^2}\right)$$

$$= r\%$$

The maximum percentage error in the diagonal is $r\%$.

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56. The legs of a right triangle are measured to be 3 cm and 4 cm, with the maximum error of 0.05 cm in each measurement. Use differentials to approximate the maximum possible error in the calculated value of (a) the hypotenuse and (b) the area of triangle.

solve

Given : $x = 3$ cm, $y = 4$ cm

$dx \leq 0.05$ cm, $dy \leq 0.05$ cm

(a) Hypotenuse : $z = \sqrt{x^2 + y^2}$, at the given point

$z = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$ cm.

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = \frac{x}{\sqrt{x^2+y^2}} dx + \frac{y}{\sqrt{x^2+y^2}} dy$$

$$dz = \frac{x}{z} dx + \frac{y}{z} dy = \left(\frac{3}{5}\right)(0.05) + \left(\frac{4}{5}\right)(0.05)$$

$$= 0.03 + 0.04 = 0.07 \text{ cm}$$ Ans

(b) Area : $A = \frac{1}{2} xy$

$$dA = \frac{\partial A}{\partial x} dx + \frac{\partial A}{\partial y} dy = \frac{1}{2} y\, dx + \frac{1}{2} x\, dy$$

$$dA = \frac{1}{2}(4)(0.05) + \frac{1}{2}(3)(0.05)$$

$$dA = 0.1 + 0.075$$

$$= 0.175 \text{ cm}^2$$ Ans

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57. The period T of a simple pendulum with small oscillations is calculated from the formula $T = 2\pi\sqrt{L/g}$ where L is the length of pendulum and g is the gravity. Suppose that measured values of L and g have errors of at most 0.5% and 0.1%, respectively use differentials to approximate the maximum percentage error in the calculated value of T.

Solve

Error in length $= \frac{dl}{l} \times 100 = 0.5\%$

Error in gravity $= \frac{dg}{g} \times 100 = 0.1\%$

Error in time period (T) $= \frac{dT}{T} \times 100 = ?$

Function: $T = 2\pi\sqrt{l/g} = 2\pi l^{\frac{1}{2}} g^{-\frac{1}{2}}$

$$dT = \frac{\partial T}{\partial l} dl + \frac{\partial T}{\partial g} dg$$

$$dT = (2\pi \cdot 1/2\, l^{-\frac{1}{2}} g^{-\frac{1}{2}})dl + (2\pi l^{\frac{1}{2}} \cdot -\tfrac{1}{2}\, g^{-\frac{3}{2}})dg$$

$$\frac{dT}{T} = \frac{(2\pi \cdot \frac{1}{2} l^{-\frac{1}{2}} g^{-\frac{1}{2}})dl}{2\pi l^{\frac{1}{2}} g^{-\frac{1}{2}}} + \frac{(2\pi l^{\frac{1}{2}} \cdot -\frac{1}{2} g^{-\frac{3}{2}})dg}{2\pi l^{\frac{1}{2}} g^{-\frac{1}{2}}}$$

$$\frac{dT}{T} = \frac{1}{2} l^{-1} dl - \frac{1}{2} g^{-1} dg$$

$$\frac{dT}{T} = \frac{1}{2}\left(\frac{dl}{l}\right) - \frac{1}{2}\left(\frac{dg}{g}\right) =$$

$$\frac{dT}{T} \times 100 = \frac{1}{2}\left(\frac{dl}{l} \times 100\right) - \frac{1}{2}\left(\frac{dg}{g} \times 100\right)$$

$$= \frac{1}{2}(0.5\%) + \frac{1}{2}(0.1\%)$$

$$= 0.3\%$$

Ans

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58. According to the ideal gas law, the pressure, temperature and volume of a confined gas are related by $P = kT/V$, where k is a constant. Use differentials to approximate the percentage change in pressure if the temperature of the gas is increased 3% and the volume is increased 5%.

Solution

$$P = \frac{kT}{V}$$

$$dP = \frac{\partial P}{\partial T}\,dT + \frac{\partial P}{\partial V}\,dV$$

$$dP = \frac{k}{V}\,dT + \left(-\frac{kT}{V^2}\right)dV$$

$$\frac{dP}{P} = \frac{k/V\,dT}{kT/V} - \frac{kT/V^2\,dV}{kT/V}$$

$$\frac{dP}{P} = \frac{dT}{T} - \frac{dV}{V}$$

$$\frac{dP}{P} \times 100 = \frac{dT}{T} \times 100 - \frac{dV}{V} \times 100$$

$$= 3\% - 5\%$$

$$= -2\%$$

There is about 2% decrease in pressure.

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60. The total resistance R of three resistances R_1, R_2, R_3 connected in parallel is given by $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$. Suppose that R_1, R_2 and R_3 are measured to be 100 ohms, 200 ohms and 500 ohms, respectively, with a maximum error of 10% in each. Use differentials to approximate the maximum percentage error in the calculated value of R.

Sol:

Formula for total resistance R in parallel.

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Given:

$$\text{Error in } R_1 = \frac{dR_1}{R_1} \times 100 = 10\%$$

$$\text{Error in } R_2 = \frac{dR_2}{R_2} \times 100 = 10\%$$

$$\text{Error in } R_3 = \frac{dR_3}{R_3} \times 100 = 10\%$$

We need to find maximum percentage error in the total resistance R, which is:

$$\frac{dR}{R} \times 100 = ?$$

$$R^{-1} = R_1^{-1} + R_2^{-1} + R_3^{-1}$$

$$d(R^{-1}) = d(R_1^{-1}) + d(R_2^{-1}) + d(R_3^{-1})$$

$$(-1)R^{-2}\,dR = (-1)R_1^{-2}\,dR_1 + (-1)R_2^{-2}\,dR_2 + (-1)R_3^{-2}\,dR_3$$

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$$\frac{1}{R^2}dR = \frac{1}{R_1^2}dR_1 + \frac{1}{R_2^2}dR_2 + \frac{1}{R_3^2}dR_3$$

Find the relative error dR/R, so Multiply the whole eqn by R.

$$\frac{R}{R^2}dR = R\left(\frac{1}{R_1^2}dR_1 + \frac{1}{R_2^2}dR_2 + \frac{1}{R_3^2}dR_3\right)$$

$$\frac{dR}{R} = R\left(\frac{1}{R_1^2}dR_1 + \frac{1}{R_2^2}dR_2 + \frac{1}{R_3^2}dR_3\right)$$

$$\frac{dR}{R} = R\left(\frac{1}{R_1}\left(\frac{dR_1}{R_1}\right) + \frac{1}{R_2}\left(\frac{dR_2}{R_2}\right) + \frac{1}{R_3}\left(\frac{dR_3}{R_3}\right)\right)$$

$$\frac{dR}{R}\times 100 = R\left[\frac{1}{R_1}\left(\frac{dR_1}{R_1}\times 100\right) + \frac{1}{R_2}\left(\frac{dR_2}{R_2}\times 100\right) + \frac{1}{R_3}\left(\frac{dR_3}{R_3}\times 100\right)\right]$$

$$= R\left[\frac{1}{R_1}(10\%) + \frac{1}{R_2}(10\%) + \frac{1}{R_3}(10\%)\right]$$

$$= R\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)(10\%)$$

$$= R\left(\frac{1}{R}\right)(10\%$$

$$= 10\%$$

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